INORGANIC CUMULATIVE EXAM

April 20, 1996

The attached page gives excerpts from a March 1996 article published in Inorganic Chemistry. Answer 9 out of the following 11 questions based on this article. It is recommended you read the entire article before you start to answer questions.

1. At the beginning of the "Synthesis" section, what is meant by "photogenerated $W(CO)_5THF$"? Why is this a logical reagent for work with a new ligand?

2. Complex **1** (described but not illustrated) is said to have "approximate C_{2v} symmetry". Explain how the complex differs from exact C_{2v} symmetry.

3. Explain why the ^{13}C NMR pattern for the carbonyls of complex **1** is a 2:1:1 pattern.

4. How can a dianion (complex **3**) be made by reacting a neutral complex ($W(CO)_5THF$) with the monoanion $[NHC_6H_4NH_2]^-$?

5. Why is complex **2** referred to as "coordinatively unsaturated"?

6. Why is complex **2** referred to as "electronically unsaturated"?

7. Complex **3** is said to have "approximate C_{3v} symmetry". Explain how the complex differs from exact C_{3v} symmetry.

8. Using the structural information of Table 2, point out the one or two bond angles which most significantly differ from ideal if the complex is considered (a) trigonal bipyramidal, or (b) square pyramidal. Indicate which two atoms you chose for the axial positions in your analysis as a trigonal bipyramidal unit, and which atom is axial in your analysis as a square pyramidal unit.

9. According to the data of this work which is a better pi donor, $-NH^-$ or $-O^-$? What data did you use to determine your answer, and how do these data support your conclusion?

10. According to the data of this work which is better for pi donation, the axial or equatorial position ? What data did you use to determine your answer, and how do these data support your conclusion?

11. The carbonyls of complex **2** were said to be fluxional at -35^oC. What does this mean? Describe what experiment was likely done at -35^oC to reach this conclusion, and the kind of data the experiment probably produced.

Coordinatively and Electronically Unsaturated Tungsten(0) Carbonyl Complexes Stabilized by π-Donating Amido Ligands

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Results

Synthesis. The reaction of photogenerated $W(CO)_5THF$ with the monodeprotonated 2-aminophenolate anion from $[Et_4N][OC_6H_4NH_2]$ carried out at ambient temperature provided in 90% yield the chelated tetracarbonyl derivative $[Et_4N][W(CO)_4OC_6H_4NH_2]$ (**1**). Formation of complex **1** was confirmed by infrared spectroscopy, as evidenced by a four-band pattern in the ν(CO) region, which is consistent with approximate C_{2v} symmetry for a *cis*-disubstituted tetracarbonyl derivative.[16] The related coordinatively unsaturated complex $[Et_4N]_2[W(CO)_3OC_6H_4NH]$ (**2**) was synthesized by the slow addition of 1 equiv of $[Et_4N][OH]$ in methanol to a THF/CH_3CN solution of **1**. This reaction was clean in that no carbonyl-containing species were produced other than the tricarbonyl tungsten dianion.

The diamido coordinatively unsaturated complex, $[Et_4N]_2[W(CO)_3NHC_6H_4NH]$ (**3**), was prepared by following the procedure previously employed for synthesizing its catecholate analog.[9,17] That is, $W(CO)_5THF$ (produced photochemically in THF) was reacted with 2 equiv of $[Et_4N][NHC_6H_4NH_2]$ over a period of 3 h. The long reaction period was required because of the low solubility of $[Et_4N][NHC_6H_4NH_2]$ in THF. However, as the reaction mixture was stirred, the amido ligand was slowly drawn into solution as the reaction occurred. This reaction is presumed to occur in a stepwise fashion with the deprotonated amine functionality first displacing the THF from the metal center. This would then be followed by additional loss of CO due to the strong π-donating ability of the amido species. Loss of CO allows the appended NH_2 ligand to bind to the metal center. Once the amine functionality is chelated to the metal, it reacts with the second equivalent of free ligand and is deprotonated, forming the dianionic species (**3**) and re-forming free 1,2-diaminobenzene.

Spectral Properties. As indicated earlier, complex **1** exhibits a four-band pattern in the carbonyl region of the infrared of the appropriate intensity pattern consistent with a *cis*-disubstituted metal center with ν(CO) bands at 1985 (w), 1848 (s), 1827 (sh), and 1794 (m) cm^{-1}. The ^{13}C NMR spectrum of **1** displays three signals for the carbonyl carbons in an approximate 2:1:1 intensity ratio.

Deprotonation of the amine functionality in complex **1** to afford the coordinatively and electronically deficient complex **2** results in the anticipated change in the ν(CO) infrared spectrum. A two-band pattern of the appropriate intensity ratio consistent with a tricarbonyl complex of *approximately* C_{3v} local symmetry was observed with bands at 1851 (s) and 1714 (vs) cm^{-1}. A summary of the ν(CO) infrared and ^{13}C carbonyl NMR data in acetonitrile for complexes **1**−**3** is provided in Table 4.

Discussion

The schematic drawings of the three closely related tungsten tricarbonyl dianions, which are coordinatively and electronically unsaturated, are provided in Figure 4, where their geometries are represented simplistically as trigonal bipyramidal. The structural parameters, which are defined in Figure 4 and summarized in Table 5, indicate the significant distortions from ideal TBP geometry observed.

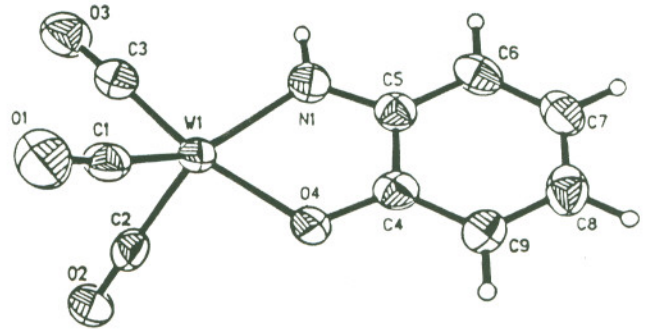

Figure 1. Thermal ellipsoid drawing of the dianion of complex **2** (50% probability) with the atomic numbering scheme.

Table 2. Selected Bond Lengths (Å)[a] and Angles (deg)[a] for $[Et_4N]_2[W(CO)_3OC_6H_4NH]$, **2**

C(1)−W(1)−C(2)	83.3(3)	C(1)−W(1)−C(3)	84.8(3)
C(2)−W(1)−C(3)	87.1(3)	C(1)−W(1)−N(1)	138.6(2)
C(2)−W(1)−N(1)	138.0(3)	(C3)−W(1)−N(1)	96.2(3)
C(1)−W(1)−O(4)	100.5(3)	C(2)−W(1)−O(4)	101.1(2)
C(3)−W(1)−O(4)	170.7(2)	N(1)−W(1)−O(4)	74.7(2)

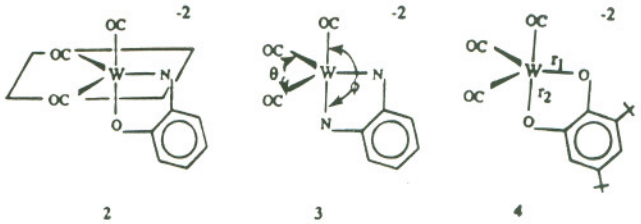

Figure 4. Schematic drawings of complexes **2**−**4** where the dianions are represented ideally as having trigonal bipyramidal geometry.

Table 4. Infrared[a] and 13CO NMR[b] Data for **1**−**3**

complex	ν(CO), cm^{-1}	$\delta(^{13}CO)$, ppm
1	1985 (w), 1848 (s), 1827 (m), 1794 (m)	206, 216, 217
2	1851 (s), 1714 (vs)	239
3	1839 (s), 1707 (vs)	241

[a] Spectra determined in CH_3CN. [b] Spectra determined in CD_3CN.

Table 5. Comparative Structural Parameters for Five-Coordinate Tungsten Tricarbonyl Dianions

complex	r_1, Å	r_2, Å	θ, deg	ϕ, deg	dev from planarity, Å
2	2.078(5)	2.143(5)	83.3(3)	170.7(2)	0.005
3	2.125(5)	2.156(5)	82.2(2)	166.4(2)	0.003
4[a]	2.059(6)	2.154(6)	85.6(6)	166.7(3)	0.026